Arjun Tschand

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Education

Duke University – Durham, NC — Expected May 2027

Bachelor of Science in Electrical and Computer Engineering and Computer Science — **GPA:** 3.83/4.00

- **Coursework**: Deep Learning (Grad), Data Structures & Algorithms, Robust Machine Learning, Natural Language Processing, Probability, Linear Algebra, Multivariable Calculus, Computer Architecture, Signals & Systems

Experience

Data Science and AI Intern, SoFi — Jun 2026 – Aug 2026

- Developed a machine learning personalization system for SoFi Money's new user activation tracker, replacing a static onboarding checklist with per-user recommended actions optimized for customer lifetime value (LTV)
- Engineered a feature dataset of 1.8 million records from 7 SQL data tables across 4 Snowflake databases spanning transaction activity, product holdings, and demographic attributes, distilled into 33 model-ready features
- Trained 15 scikit-learn gradient boosting models predicting product adoption probability (avg AUC 0.86) and 15 propensity score matching models isolating LTV lift (avg SMD 0.06), combining to rank actions by EV per user
- Launched a production A/B test with the personalized activation tracker, currently live for ∼6,000 new users daily

Machine Learning Researcher, Duke University – Zavlanos Lab — Aug 2025 – Present

- Built PyTorch training pipelines applying distributionally robust optimization (GroupDRO) over excess loss, improving worst-group performance by targeting underserved groups over those inherently harder to predict
- Designed a representation learning architecture with per-group encoders and a shared classifier for heterogeneous feature spaces, aligning latent spaces via Gaussian and Wasserstein distance to improve cross-group generalization
- Evaluated on three tabular/imaging medical datasets (NHANES, Fed-Heart, EMBED), cutting worst-group loss up to 79% and excess loss up to 95% over published baselines with up to 18× fewer parameters
- Co-first author on paper submitted to International Conference on Artificial Intelligence & Statistics (AISTATS)

Software Engineering Intern, Bloomberg LP — Jun 2025 – Aug 2025

- Developed a Large Language Model (LLM)-powered diagnostic system to automate post-trade issue resolution for fixed income trade allocations, reducing resolution time by 85% on the Sell Side Specialty Assets Team
- Architected a Python and FastAPI backend with LangGraph-based agents that queries 1,200+ trade audit logs per request and leverages Bloomberg's LLMs to identify trade allocation pipeline issues and recommend resolutions
- Designed a React and TypeScript chatbot UI with conversational memory and enterprise SSO authentication
- Implemented agent observability, logging, and evaluation to monitor LLM response quality and system reliability

Software Engineering Intern, Bloomberg LP — Jun 2024 – Aug 2024

- Developed observability tooling for Bloomberg Terminal equity index workflows on the Index Reference Data Team
- Built real-time dashboards visualizing equity index usage metrics of 20,000+ daily users to monitor migration
- Authored flexible Grafana and Humio data queries to power telemetry requests across two internal data systems
- Drove adoption of the new EQIX workflow for 625+ engineers by highlighting legacy-function usage in dashboards

Projects

Crypto Options Pricing & Risk Engine — Apr 2026

- Built BTC/ETH pricer (Black-Scholes, Monte Carlo, Heston), 8.4% MAPE validated on 1,505 Deribit contracts
- Calibrated Heston via differential evolution, outperforming Black-Scholes by 166 pp in 2020/2022 crash regimes
- Designed algorithm to flag 110 contracts with >15% model-vs-market mispricings, generating trading watchlist

Realtime Drowning Detection — Jan 2024 – Jun 2024

- Built an embedded Raspberry Pi–based system for poolside drowning detection by training a real-time TensorFlow CNN classifier, achieving 96% accuracy on a custom labeled image dataset (drowning, swimming, empty pool)
- Developed a Flask and SQL backend and a Swift iOS app streaming live video with sub-2 second drowning alerts
- Self-filed USPTO provisional patent titled *Intelligent Audio-Visual Observation System to Detect Pool Drownings*

RecyclAIble — Sept 2024

- Collaborated to create computer vision-powered mechanical waste bin, reaching 95.4% classification accuracy
- Won "Best Hardware Hack" out of 62 teams at PennApps XXIII Hackathon

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, Swift, HTML/CSS
Frameworks/Tools: PyTorch, TensorFlow, React, Pandas, NumPy, LangGraph, FastAPI, AWS, Git, Docker, Linux